

Computer Networks

Season 2024-III

Workshop No. 2 — Sockets and Services, final report.

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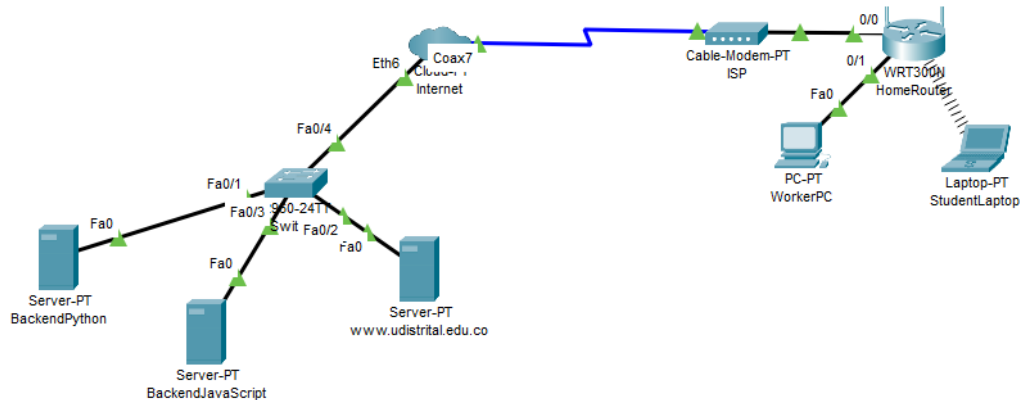


Figure #1, “topological diagram”.



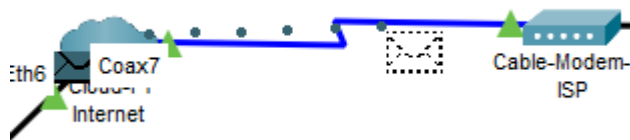
Figure #2, “www.udistrital.com with services”.

Network Decisions:

The previous workshop was used to simulate the client-server relationship between the student machine and the website, complementing with the new backend servers and a switch for the ISP connection in the cloud, they were maintained the DHCP services and agreed new functionalities for the website.

OSI Model:

- Physical Layer



Ethernet connections and serial connections brings the médium for transmits.

- Data Link Layer

PDU Information at Device: Switch0

[OSI Model](#) [Inbound PDU Details](#) [Outbound PC](#)

At Device: Switch0
Source: www.udistrital.edu.co
Destination: WorkerPC

In Layers	Out Layers
Layer7	La
Layer6	La
Layer5	La
Layer4	La
Layer3	La
Layer 2: Ethernet II Header 0040.0B50.D98E >> 000A.4184.5201	La 00
Layer 1: Port FastEthernet0/2	La

1. FastEthernet0/2 receives the frame.

- Network Layer:

In Layers	Out Layers
Layer7	Layer7
Layer6	Layer6
Layer5	Layer5
Layer4	Layer4
Layer 3: IP Header Src. IP: 193.168.100.200, Dest. IP: 192.168.0.102 ICMP Message Type: 0	Layer3
Layer 2: Ethernet II Header 0007.EC5A. 0347 >> 0001.63AE.6C61	Layer2
Layer 1: Port FastEthernet0	Layer1

1. The packet's destination IP address matches the device's IP address or the broadcast address. The device de-encapsulates the packet.
2. The packet is an ICMP packet. The ICMP process processes it.
3. The ICMP process received an Echo Reply message.
4. The Ping process received an Echo Reply message.

- Transport and session layer:

These layers don't show explicit at packet tracer, but the use of ports and open sessions is important.

- Presentation and application layer:

